Finite Element Analysis of 6061-T6 Aluminum Tensile Specimens With and Without a Central Hole

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This report presents a finite element analysis (FEA) study of ASTM-standard 6061-T6 aluminum tensile test dogbone specimens using Ansys Static Structural simulations. Both two-dimensional and three-dimensional models are considered – all with the same outer dimensions – including specimens with and without a central circular hole to investigate stress concentration effects. Plastic deformation was modeled, using a multilinear kinematic hardening law based on 6061-T6 aluminum alloy plastic true stress-strain data. The objectives of this study are to identify the maximum boundary conditions (namely displacement for the 2D tests and load for the 3D test) at the onset of failure, evaluate the mesh convergence of the solutions, quantify stress concentration behavior in the elastic and plastic regimes, and evaluate the residual deformation from loading the specimen to near-failure and then unloading it. By comparing the numerical results of the 2D tensile test simulations, the peak reaction force was found to serve as a robust indicator of failure in FEA, that quickly converges and is less sensitive to numerical instability than local measures like displacement and stress. With this failure indicator, the specimens failed at imposed boundary conditions of 5.308 mm, 0.709 mm, and 13.580 kN, for the 2D specimen without hole, 2D specimen with central hole, and 3D specimen respectively. For the 2D specimen with the central hole, the Elastic stress concentration factors calculated from the simulation results closely matched the theoretically predicted value of $K_t \approx 2.321$ in the elastic regime, and then decrease after yielding due to plastic stress redistribution. The three-dimensional simulation further revealed through-thickness stress variations and residual deformation that were not captured by two-dimensional models. Lastly, by simulating the loading and unloading of the 3D specimen to and from 96% of its FEA predicted ultimate tensile load, the specimen was found to have permanently elongated 0.704 mm.

Nomenclature

A = initial gross cross-sectional area of the specimen [mm^2]

D = specimen width [mm]

d = hole diameter [mm]

E = Young's modulus, GPa

F = force [N]

h = mesh element size [mm]

K_t = stress concentration factor [−]

L = instantaneous specimen length, mm

L_0 = initial specimen length, mm

t = thickness [mm]

δ = displacement [mm]

ϵ = strain [−]

σ = stress [Pa]

I. Introduction

A tensile test is a fundamental mechanical test that involves pulling a material sample to characterize its stress-strain response under uniaxial loading. This study focuses on analyzing the mechanical behavior of 6061-T6 aluminum specimens (with and without a central circular hole stress concentration) through tensile tests, following ASTM standards. The ASTM standard that governs the tensile testing of metallic materials is ASTM E8 / E8M — Standard Test Methods for Tension Testing of Metallic Materials (Ref. [1]). (The corresponding standard for plastics would be ASTM D638 — Standard Test Method for Tensile Properties of Plastics (Ref. [2]).) While ASTM is the most common in the United States, several other international and national organizations maintain high-authority standards for the tensile testing of metals. Perhaps the most frequently used alternative to ASTM E8 is ISO 6892-1 (Metallic materials — Tensile testing — Part 1: Method of test at room temperature) from the International Organization for Standardization (Ref. [3]). While the geometries of the tensile test dogbone specimen varies slightly between ISO and ASTM standards, the fundamental physics and analysis remains comparable across both standards.

Aluminum 6061-T6 is a common structural aluminum alloy. It is widely used in the aerospace industry for structural components throughout the fuselage, wings, and landing gear, and in the marine engineering industry for ship and submarine frames — just a few of its many applications. Its popularity can be attributed to key advantages that aluminum alloys have over steel, such as their densities being approximately one-third as dense as steel (a compelling advantage for vehicles where fuel efficiency is a concern), and the fact that aluminum naturally forms an oxide layer that protects it against further corrosion – whereas steels typically require additional processes like galvanization or painting to achieve similar protection.

Throughout this study, it will be important to distinguish between true and engineering stress and strain. Engineering

stress assumes that the cross-sectional area of a specimen remains the same throughout a test, whereas true stress accounts for the fact that a specimen becomes thinner as it is stretched.

$$\sigma_{\text{engineering}} = \frac{F}{A_0} \tag{1}$$

$$\sigma_{\text{true}} = \frac{F}{A} \tag{2}$$

Moreover, engineering strain is defined as the change in length divided by the original length of the specimen. True strain, on the other hand, is the integral of the fraction of changes in the sample's length divided by the sample's (instantaneous) length, which results in a logarithmic relationship between true strain and the fraction of the length of the sample divided by its original length.

$$\epsilon_{\text{engineering}} = \frac{L - L_0}{L_0} = \frac{\Delta L}{L_0} \tag{3}$$

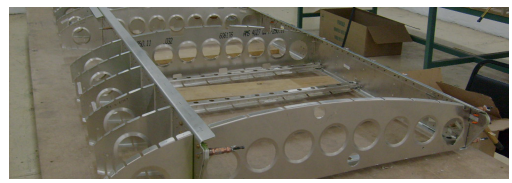
$$\epsilon_{\text{true}} = \ln\left(\frac{L}{L_0}\right) \tag{4}$$

Additionally, elasticity refers to the ability of a material to return to its original size and shape upon the removal of an applied load that deformed it, whereas plasticity describes irreversible deformations of materials, caused by applied loads that exceed the material's yield strength. Fatigue life is a measure of the number of loading cycles a specimen can withstand before failing. In industry, the fatigue life of a specimen is typically measured using cyclic loading machines that repeatedly load and unload it until failure occurs. The data collected from these tests is then typically used to generate a graph of the cyclical stress and number-of-cycles at which the specimen fails (commonly referred to as an S-N curve). Knowing the fatigue life of a specimen is critical for its extended operation in applications such as electronic components, where solder joints, interconnects, and circuit boards must endure constant expansion and contraction throughout a device's thermal cycles, as well as in turbine blades for jets and ships that must withstand millions of centrifugal loading cycles. The tensile tests simulated in this study model non-cyclic elastic and plastic deformations of aluminum 6061-T6 specimens, while offering insights into residual deformation and its role in material fatigue.

In a physical tension test, failure is defined by the fracture or rupture of the specimen. However, in a Static Structural Ansys simulation, the software does not inherently model the physical separation of the geometry into distinct pieces. Instead, failure is identified through indicators such as a concave-down inflection point in the force-displacement curve—signaling a loss of load-carrying capacity, and localized peaks in equivalent plastic strain. These signals indicate that the material has reached its limit for plastic deformation and can no longer structurally support the applied load. Sometimes, it can even be identified simply by the program's ability to converge to a solution beyond a certain imposed boundary condition.

One of the specimens studied in this report has a hole in the middle. The importance of understanding the stress concentration factor (K_t) around a hole extends well beyond fastener (screws/nails/rivet) holes. For instance, holes are regularly cut out of structures such as gym equipment to facilitate adjustable configurations — such as the adjustable-height squat rack shown in Figure 1a. Ensuring that these structures can support heavy weights is crucial to securing the user's safety. Additionally, structures are frequently holed out for light-weighting, as exemplified by the lightening holes in the spars of the aircraft wing in Figure 1b. Understanding the stress-concentrations of the lightening holes under the intense loads exerted on the wings during flight is critical to the plane's functionality.

(a) Gym rack with holes for height adjustment (Ref. [4]).



(b) Lightening holes in aircraft wing spars (Ref. [5]).

Fig. 1 Holed Structure Examples.

Ultimately, this study seeks to evaluate the mechanical behavior of 6061-T6 aluminum through a three-stage Finite Element Analysis (FEA) simulation arc: a 2D baseline, a 2D stress concentration analysis, and a 3D investigation of out-of-plane effects.

II. Methods

A. Software

All of the modeling and FEA for this study was conducted using Ansys Student 2025 R2 [6]. All contour plots in this study were generated with Ansys, whereas the plots were all generated in MATLAB R2021a [7] using the relevant raw simulation data exported from Ansys.

B. Material Model

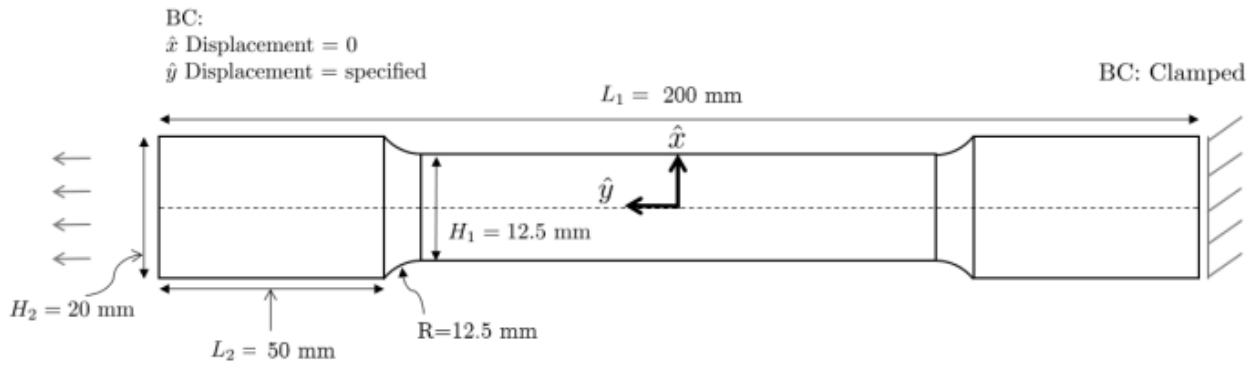
All simulations used 6061-T6 aluminum with linear elastic properties defined by Young's modulus, Poisson's ratio, and density corresponding to room temperature conditions (22°C) – 689 MPa, 0.33, and 2700 kg m^{-3}, respectively. Plastic deformation was modeled using a multilinear kinematic hardening law defined by a table of empirical the material's true plastic stress-strain data [1].

Table 1 6061-T6 aluminum alloy plastic true stress-strain data.

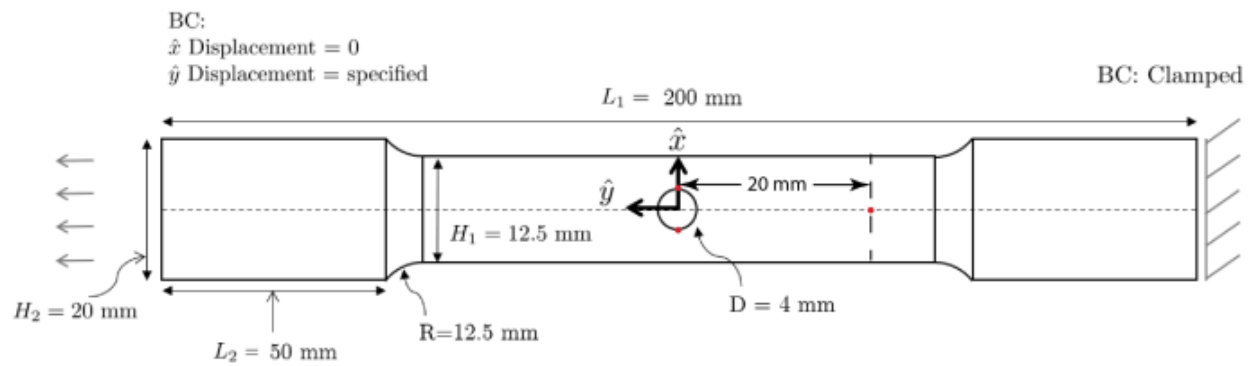
Plastic Strain (m m^{-1})	Stress (MPa)	Stress Points
0	287.19	Yield Point
0.0001	291.78	—
0.0011	301.64	—
0.0027	313.61	—
0.0073	320.64	—
0.011	330.63	—
0.0153	340.63	—
0.0156	346.73	—
0.0116	342.73	—
0.0454	360.16	—
0.0556	369.07	—
0.1009	374.07	—
0.1189	374.9	—
0.1351	375.25	Rupture Point

C. Geometry and Boundary Conditions

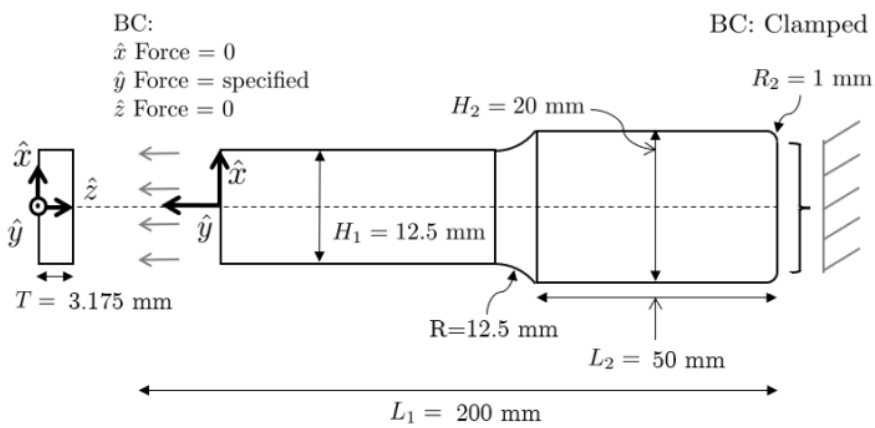
Two ASTM-standard dogbone specimens were modeled in 2D – one with a central 4 mm diameter hole [2b] and one without [2a]. A third dogbone was modeled in 3D [2c]. The thicknesses of the 2D models were entered directly into the FEA software, as 3.175 mm for both, so that all three models have the same outer dimensions. All models were created using Ansys Design Modeler.



(a) 2D tensile test specimen (without hole).



(b) 2D tensile test specimen with central 4 mm diameter hole. The three red points mark the locations of the stress probes used.



(c) 3D tensile test specimen.

Fig. 2 Tensile test dogbone specimen geometries.

In all simulations, one end of the specimen was fully constrained. For the 2D simulations, displacement-controlled loading was applied to other end of the specimen. For the 3D simulation, force-controlled loading was applied to the other end (flat face) of the specimen.

D. Meshing Strategy

All models were meshed using the Ansys mesh *Face Sizing* feature on the $\hat{x} - \hat{y}$ plane of the Dogbone specimen. For the 2D model with the central hole, the mesh was further refined near the hole boundary using the *Edge Sizing* feature on the hole edge to more accurately capture stress concentrations. For the 2D meshes, the behavior of the *Face Sizing* feature was toggled from “Soft” to “Hard” to prioritize it over other constraints. For the 3D model, a sweep method was used, with an "All Quad" *Free Face Mesh Type* and a *Sweep Num Divs* of 4, to sweep the face mesh through the body and ensure that 4 elements existed along its thickness ($\hat{z}$ direction in Figure 2c).

Mesh refinement studies were conducted in the 2D simulations to assess solution convergence. For the mesh of the 2D specimen without the hole, the element size of the mesh was incrementally reduced to progress from a coarse mesh to a fine one. For the 2D specimen with the central hole, the same simulation was carried out using two different mesh element sizes: 2 mm (coarse) versus 0.5 mm (fine). For the 3D model, the mesh element size was adjusted to target approximately 16,000 mesh nodes. An element size of 2.15 mm was settled upon, resulting in a mesh with 16,171 nodes.

E. Solution Procedure

Large Deflection was enabled in all simulations to study the effects of plastic deformations on the specimens. The boundary condition was defined at 5 timesteps using "Tabular Data" for all of the simulations, except in the case of loading and unloading the 3D specimen, for which 5 timesteps were used to load the specimen and then and additional 5 to unload it (10 timesteps total). Additionally, *Automatic Time Stepping* (with 10 initial substeps, 10 minimum substeps, and 100 maximum substeps) was used to interpolate between the manually defined timesteps and incrementally apply the non-fixed boundary condition.

III. Results

A. Two-dimensional nonlinear material response

A 2D tensile test on the dogbone specimen without a hole [2a] was first simulated using a coarse mesh—with an element size of 7.17 mm. With this coarse mesh, failure was first observed via the presence of a peak reaction force (concave-down inflection point) of 13.616 kN at a displacement of 5.643 mm [3].

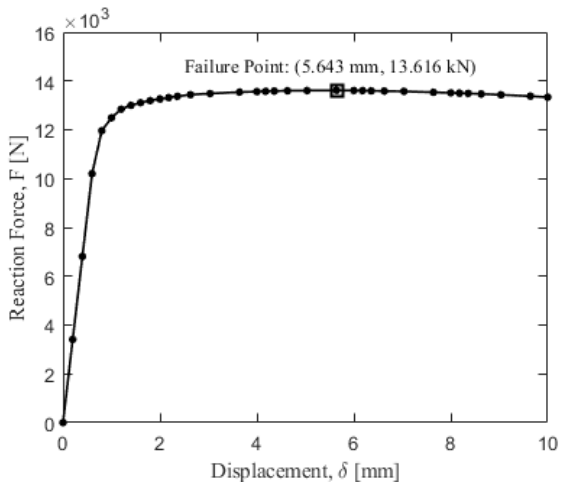
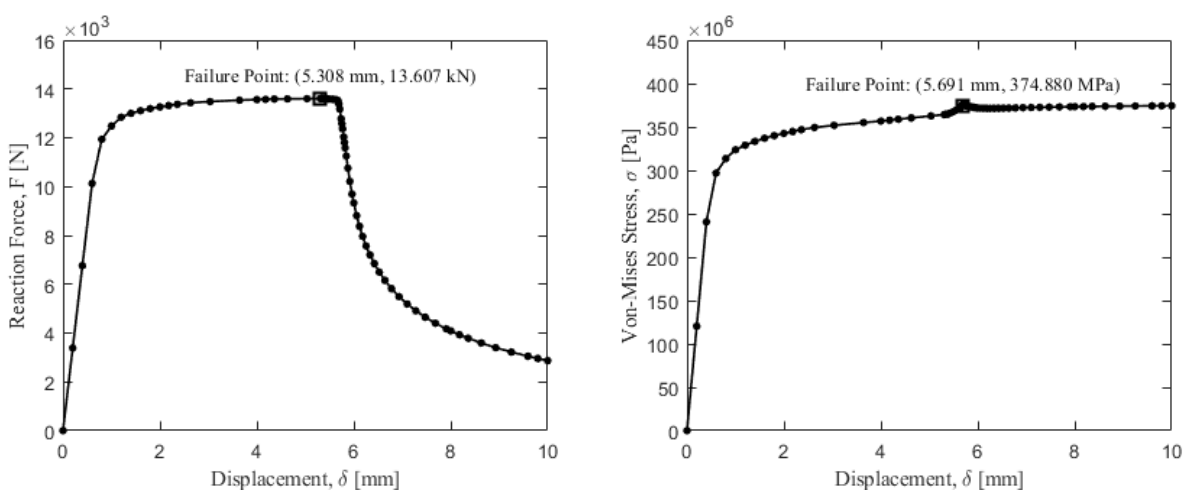


Fig. 3 Reaction force vs. boundary displacement condition (Coarse mesh, h = 7.17 mm).

Past this failure point, the boundary reaction force drops off, signaling that the cross-sectional area of the specimen decreases faster than the material hardens, causing its total load-bearing capacity to drop.

With the mesh refined to an element size of 0.50 mm, this same failure indicator would suggest that failure would occur a peak reaction force of 13.607 kN at a displacement of 5.308 mm [4a]. An alternative failure indicator is a global peak followed by a plateau in the maximum von-Mises stress of the specimen. This signals that the material has reached its ultimate/breaking stress. Using this indicator, failure is predicted to occur at a displacement of 5.691 mm, with a maximum von-Mises stress of 374.880 MPa [4b].



(a) Reaction force vs. boundary displacement condition. **(b) Von-Mises stress vs. boundary displacement condition.**

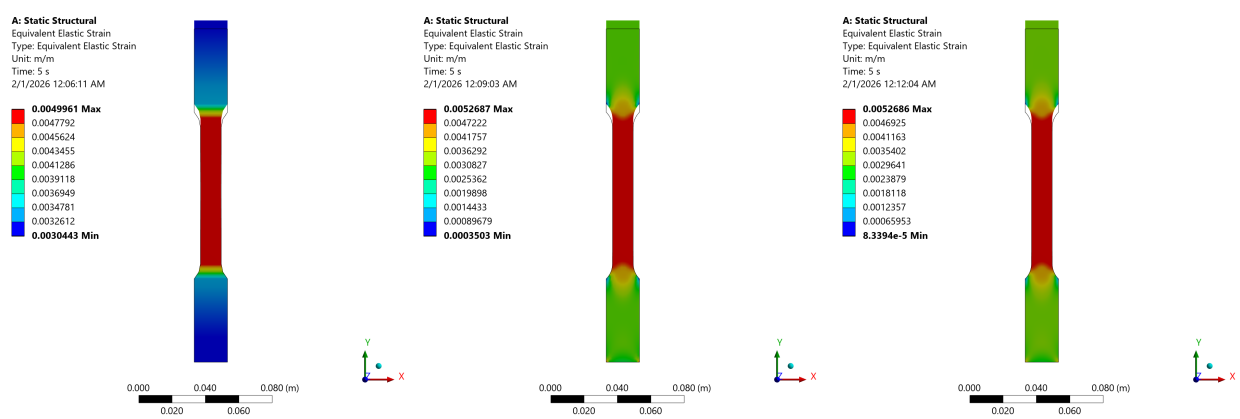
Fig. 4 Fine mesh (h = 0.50 mm).

Note that changing the mesh element size from 7.17 mm to 0.50 mm changed the displacement at which failure was predicted to occur by about 6%, whereas the force at which failure was predicted to occur changed by less than 0.1%,

suggesting that predictions of the displacement at which failure will occur are more sensitive to the resolution of the mesh than predictions of the force reaction at failure.

Although local stress peaks and global reaction force can both serve as failure indicators, the peak in the force-displacement curve is the more robust metric for identifying the ultimate structural capacity. Unlike local stress values at individual nodes – which are highly sensitive to mesh density and susceptible to numerical noise – global quantities such as the reaction force are governed by the overall stiffness and integrated deformation of the structure. These global indicators typically converge more rapidly with mesh refinement, providing a more stable and mesh-independent representation of the specimen's behavior. For this analysis, the failure point is identified at a maximum displacement of 5.308 mm, corresponding to the peak reaction force (Ultimate Tensile Strength) obtained from the fine-mesh simulation. This approach ensures that the predicted failure is a result of structural instability and plastic localization rather than localized numerical artifacts at the element level.

The simulation was rerun with a maximum boundary displacement condition of 5.096 mm – which is 96% of the maximum displacement before failure (used because FEA can be prone to solution convergence issues at or near failure) – three times, each with a different mesh element size: 7.17 mm (coarse), 2 mm (refined), and 0.5 mm (fine). The results are in Figure 5.



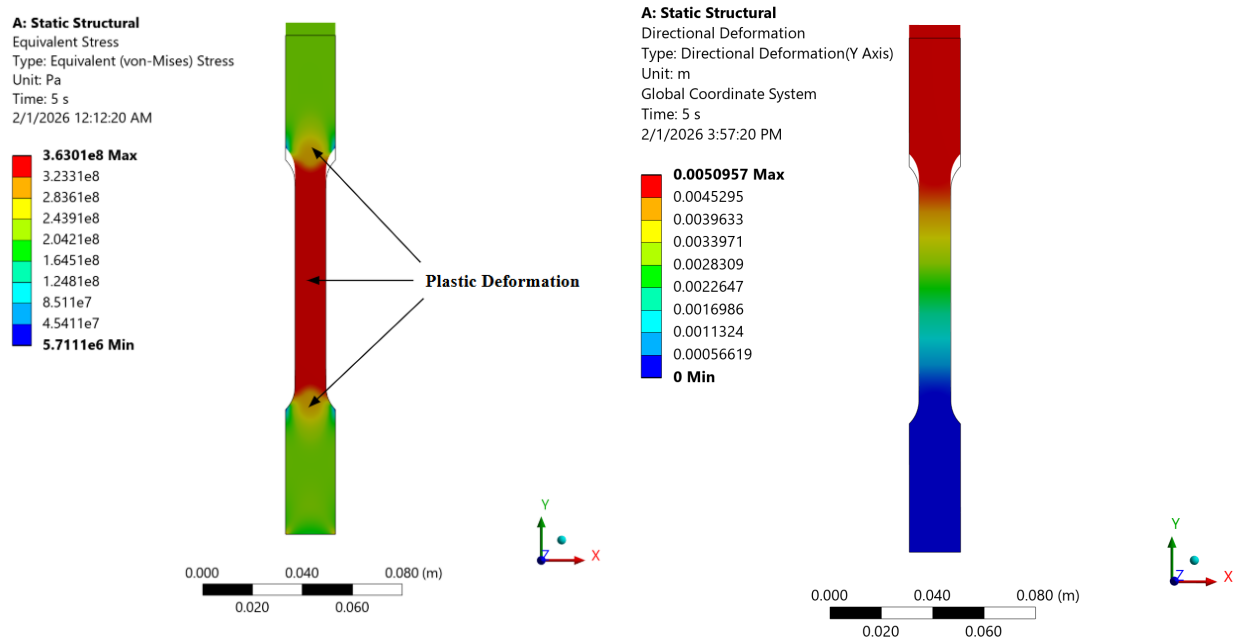
(a) Coarse mesh (h = 7.17 mm). **(b) Refined mesh (h = 2.00 mm).** **(c) Fine mesh (h = 0.50 mm).**

Fig. 5 Contours for von-Mises strains on the X-Y plane at 96% of the maximum displacement.

With the refined mesh, the simulation captures the variations in strain along $\hat{x}$ near the curved faces as well as the stress concentrations at sharp corners of the specimen [5b], whereas with the coarse mesh the simulation could not capture these finer details. The difference in the results from the refined mesh to the fine mesh were not as pronounced, with some slight adjustments to the contour legend's limits, suggesting that the mesh element size is small enough to capture the major stress variations in the specimen, and decreasing it further would yield diminishingly minute improvements.

The max von-Mises stress [6a] and displacement field contours [6b] were also plotted for this simulation (with the

fine mesh) to reveal the regions of the specimen that plastically deformed and the local displacement throughout the specimen. Note that plastic deformation is any region of the specimen with a max von-Mises stress higher than the yield point of 287.19 MPa (see Table 1).



(a) Plastic deformation indicted on von-Mises stress contour.

(b) Displacement field (directional deformation).

Fig. 6 Deformation of 2D dogbone specimen (no hole).

These results indicate that the entire narrow section of the dogbone undergoes plastic deformation [6a]. The displacement of the specimen along $\hat{y}$ varies linearly between the two boundary displacement conditions: 0 mm (fixed end) to 5.096 mm (forced displacement).

Furthermore, the results imply that failure will most likely initiate near the middle of this narrow section of the dogbone, $y = 100$ mm, where plastic deformation accumulates. Figure 5 illustrates that plastic strain is localized to this narrow region of the dogbone across mesh refinements, indicating that this region will be the first to undergo permanent deformation and, eventually, rupturing.

B. Stress concentration analysis

The 2D tensile test of the specimen with a central hole was first conducted using a coarse mesh with an element size of 2 mm. The displacement at which failure was predicted to occur was found to be around 0.709 mm, using the same force-reaction peak indicator that has been described. The far-field von-Misesstress measurements, $\sigma_{\text{far, actual}}$, collected by the probe located away from the hole [2b] were sanity-checked against the predicted far-field stress induced by the

reaction force, F, on the dogbone, $\sigma_{\text{far, theory}}$:

$$\sigma_{\text{far, actual}} \approx \sigma_{\text{far, theory}} = \frac{F}{A} \tag{5}$$

$$A = D * t \rightarrow A = (12.5 \text{ mm}) * (3.175 \text{ mm}) = 46.875 \text{ mm}^2 \tag{6}$$

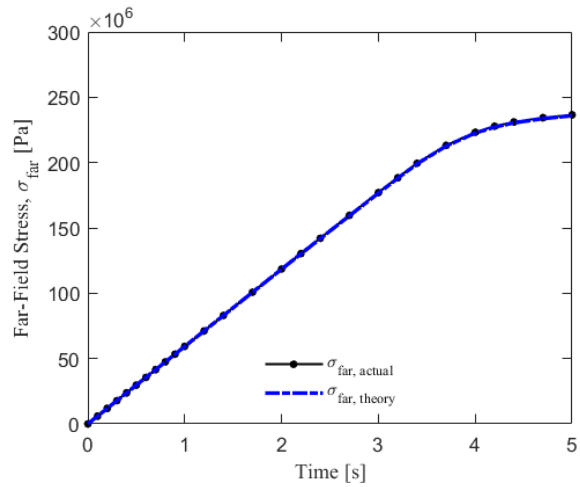


Fig. 7 Far-field stress in the 2D specimen with central hole (coarse mesh).

The plots of $\sigma_{\text{far, actual}}$ and $\sigma_{\text{far, theory}}$ almost perfectly overlap each other, which validated the simulation setup.

The far-field normal stress, σ_{yy}^{far}, was also collected at this probe location away from the hole. Meanwhile, the maximum normal stress at the hole boundary, σ_{yy}^{max}, was computed for each timestep by taking the mean of the normal stresses measured by the two probes on the hole boundary, $\sigma_{yy}^{\text{max, 1}}$ and $\sigma_{yy}^{\text{max, 2}}$ [2b].

$$\sigma_{yy}^{\text{max}} = \frac{\sigma_{yy}^{\text{max, 1}} + \sigma_{yy}^{\text{max, 2}}}{2} \tag{7}$$

The nominal stress near the hole (σ_{nom}), and the stress concentration factor (K_t) were then computed:

$$\sigma_{\text{nom}} = \frac{F}{(D-d)t} \tag{8}$$

$$K_t = \frac{\sigma_{yy}^{\text{max}}}{\sigma_{\text{nom}}} \tag{9}$$

As a secondary validation, an independent estimate of K_t was computed using the relation:

$$\frac{\sigma_{yy}^{\text{max}}}{\sigma_{yy}^{\text{far}}} = K_{t,\text{ check}} \left(\frac{1}{1 - d/D} \right) \rightarrow K_{t,\text{ check}} = \frac{\sigma_{yy}^{\text{max}}}{\sigma_{yy}^{\text{far}}} (1 - d/D) \tag{10}$$

Then, the theoretical K_t was computed using the polynomial fit formula for the stress concentration factor around a central hole in a finite-width thin plate [8]:

$$K_t = 3.00 - 3.140\left(\frac{d}{D}\right) + 3.667\left(\frac{d}{D}\right)^2 - 1.527\left(\frac{d}{D}\right)^3 \tag{11}$$

The result was $K_{t,\text{theory}} \approx 2.321$.

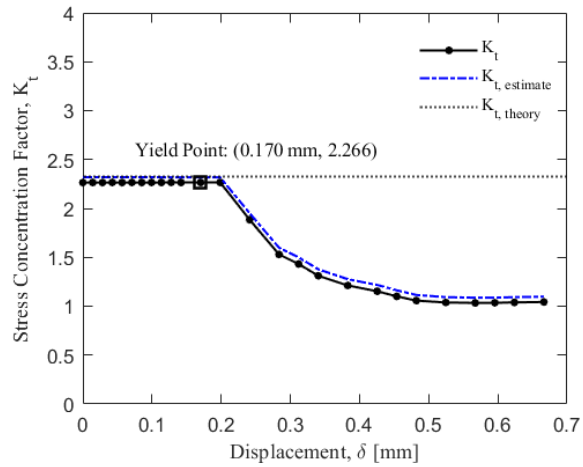
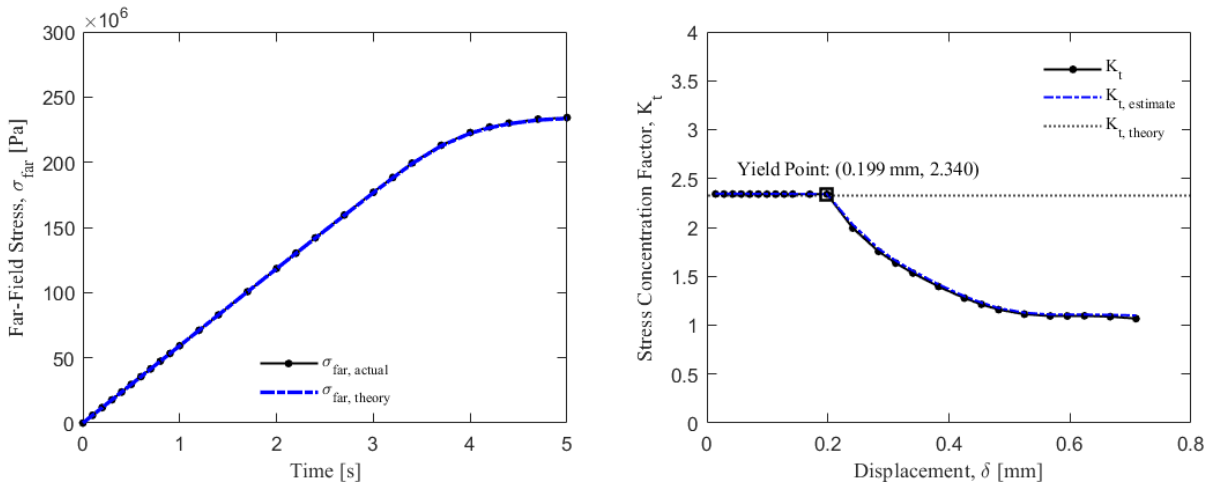


Fig. 8 Stress concentration factor vs. displacement (coarse mesh).

As shown in Figure 8, in the elastic regime, the K_t values calculated from the simulation data, with both methods, were close to (within 3% error of) the theoretically predicted value. Past the yield point, the stress concentration factor drops off – rapidly at first, and then it gradually plateaus. This is because, as the specimen is incrementally pulled in tension, the induced stress is initially relatively concentrated around the hole boundary, leading this region to yield before the rest of the material. Once this region starts to plastically deform, it also kinematically hardens. As a result, any further stress induced in the specimen stress is increasingly redistributed to nearby areas which have not yielded or at least have not hardened as much. This redistribution reduces the fraction of the total load carried by the hole boundary, resulting in a decreasing effective stress concentration factor, until eventually, segments of material spanning the full width of the narrow section of the dogbone (in the vicinity of the central hole) are also plastically deforming. Ultimately, the specimen experiences necking, and subsequently ruptures.

This full analysis was conducted again for a simulation using a fine mesh with an element size of 0.5 mm. The resulting plots are in Figure 9.



(a) Far-field stress. **(b) Stress concentration factor vs. displacement.**

Fig. 9 2D specimen with central hole – fine mesh results.

In the elastic regime, the K_t values calculated from the simulation data are now even closer to the theoretically predicted value (within 1% error). The hypothesis for why K_t then diverges in the plastic regime, discussed for the simulation with the coarse mesh, still applies, as this behavior remains consistent between the two simulations. These results suggest the far-field stress and trends in the stress-concentration factor are not particularly sensitive to the mesh element size.

C. Three-dimensional elastic–plastic analysis

The failure point for the 3D tensile test of the dogbone specimen was first estimated using the material's ultimate stress [1] and the area of the specimen:

$$(375.25 \times 10^6 \text{ Pa}) * (12.5 \text{ mm}) * (3.175 \text{ mm}) \approx 14.9 \text{ kN} \tag{12}$$

To refine this estimate, the simulation was run with a loading condition up to 15 kN. Using the peak force indicator, the rupture force of the dogbone was identified to be approximately 13.58 kN [10].

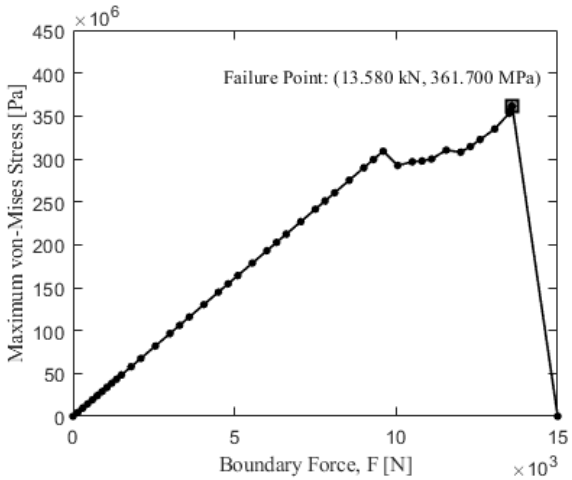
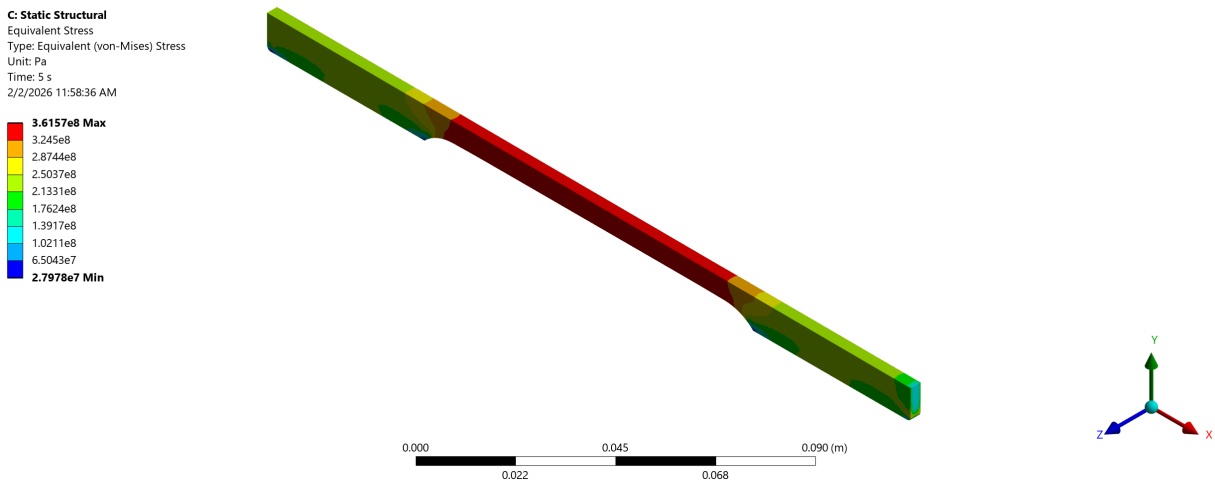
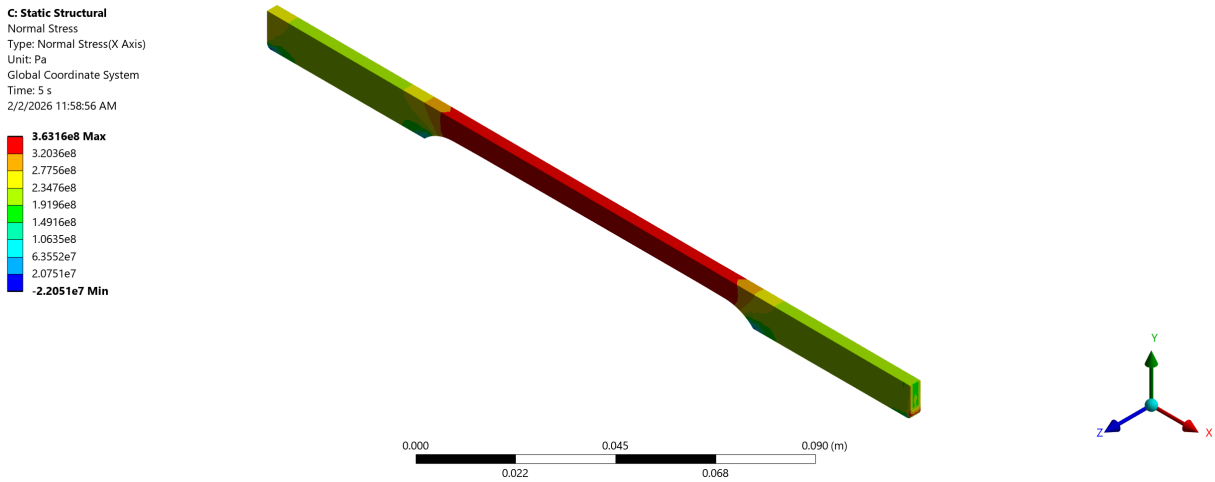


Fig. 10 Von-Mises stress vs. boundary force condition.

At the rupture load, contour plots of the von-Mises stresses and out-of-plane stress were generated in a cross-section along the $\hat{y}$ axis (refer to Figure 2c). These contour plots are shown in Figure 11.



(a) Von-Mises stress.



(b) Out-of-plane stress.

Fig. 11 3D tensile test – contour plots of stress in $\hat{y}$ cross-sction view.

The von-Mises stresses vary along the thickness near the middle of the specimen, throughout its length (note: enabling contour bands in Ansys helps to visualize this) [11a]. This variation is most pronounced near the fixed boundary face of the specimen, where the stress decreases toward the center of the face. Similarly, along the cross-sectioned face, the stress is shown to increase from the center of the face toward the outer edges. This matters in a tensile test because if the stress is higher on the outer surface, it suggests that the crack will initiate from the outer surface and then propagate inward.

The normal stress contour plot shows that this stress is tensile throughout the cross-sectioned face (only near the sharp edges at the ends of the curved faces – parallel to the $\hat{z}$-axis – were there regions of tensile stress) [11b].

A simulation of the specimen being loaded from 0 N to 13.037 kN (96% of the FEA-predicted rupture load), and then unloaded back to 0 N was then run, and the resultant boundary force vs. displacement curve is shown in Figure 12.

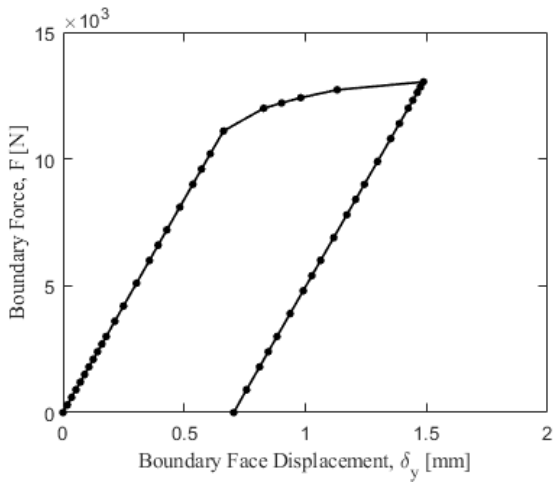


Fig. 12 Force vs. displacement of boundary face.

In its elastic region, the boundary face of the specimen displaces about 0.662 mm. It then plastically deforms to a maximum displacement of about 1.487 mm. Lastly, as the specimen is unloaded back to 0 N, its boundary face elastically deforms back to 0.704 mm. The formation of this residual deformation offers insights into the fatigue life of materials. Over time, spikes in the local stress of a material (caused by stress concentrations, sudden and/or intense loads, etc) can lead to plastic deformation. More regions of the material gradually encounter plastic deformation due to stress redistribution (as discussed in the results for 2D specimen with the hole). The material continually accumulates the resultant residual deformation until, eventually, it encounters a stress it can no longer withstand, and ruptures.

IV. Conclusion

This study investigated the elastic–plastic tensile response of 6061-T6 aluminum dogbone specimens using nonlinear finite element analysis in ANSYS. Two-dimensional baseline models, two-dimensional stress concentration models with a central circular hole, and a three-dimensional specimen were analyzed to assess failure behavior, mesh sensitivity, and stress redistribution.

Mesh refinement studies demonstrated that peak reaction force is a robust indicator of structural failure in FEA simulations of tensile tests. For specimens without a hole, failure was consistently predicted at a reaction force of approximately 13.6 kN, with displacement predictions exhibiting greater sensitivity to mesh resolution. In specimens containing a central hole, stress concentration factors closely matched the theoretically predicted value, and decreased after yielding due to plastic redistribution of stress around the hole boundary.

Three-dimensional simulations revealed through-thickness variations in von-Mises and out-of-plane stresses that are not captured in two-dimensional models. Load–unload simulations further demonstrated the accumulation of residual deformation following plastic yielding, highlighting a key mechanism by which localized plasticity can contribute

to fatigue damage over repeated loading cycles. In this particularly simulation, the 3D specimen was permanently deformed to be 0.704 mm longer than its initial length after just one load-unload cycle (to-and-from 96% of its ultimate tensile load).

Overall, the results underscore the importance of mesh refinement, appropriate failure metrics, and three-dimensional modeling when predicting plastic deformation, stress concentration behavior, and residual strain in aluminum structural components. These findings provide a foundation for future experimental validation and Digital Image Correlation studies.

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